

Status Report #3

2025-10-09

Setup

```
# Load in libraries
library(MASS)
library(e1071)

# Load data and view it (make sure everything looks right)
HeartAttack <- read.csv("HeartAttack.csv")
head(HeartAttack)
```

##	Age	Gender	Cholesterol	BloodPressure	HeartRate	BMI	Smoker	Diabetes
## 1	31	Male	194	162	71	22.9	0	1
## 2	69	Male	208	148	93	33.9	1	1
## 3	34	Female	132	161	94	34.0	0	0
## 4	53	Male	268	134	91	35.0	0	1
## 5	57	Female	203	140	75	30.1	0	1
## 6	41	Male	158	154	72	38.7	0	1
##	Hypertension	FamilyHistory	PhysicalActivity	AlcoholConsumption	Diet			
## 1	0	0	6		0	Unhealthy		
## 2	0	0	1		2	Unhealthy		
## 3	1	1	1		3	Healthy		
## 4	1	0	6		0	Healthy		
## 5	0	0	4		1	Moderate		
## 6	0	1	3		2	Moderate		
##	StressLevel	Ethnicity	Income	EducationLevel	Medication	ChestPainType		
## 1	1	Hispanic	64510	High School	Yes	Typical		
## 2	6	Asian	91773	College	No	Atypical		
## 3	3	Black	173550	College	No	Non-anginal		
## 4	3	Hispanic	43861	High School	Yes	Atypical		
## 5	1	Hispanic	83404	High School	Yes	Typical		
## 6	4	Other	113011	High School	Yes	Non-anginal		
##	ECGResults	MaxHeartRate	ST_Depression	ExerciseInducedAngina	Slope			
## 1	ST-T abnormality	173	0.52		Yes	Downsloping		
## 2	LV hypertrophy	189	3.79		Yes	Upsloping		
## 3	Normal	122	0.17		Yes	Upsloping		
## 4	ST-T abnormality	104	0.67		Yes	Flat		
## 5	ST-T abnormality	126	5.00		Yes	Flat		
## 6	LV hypertrophy	155	4.30		No	Downsloping		
##	NumberOfMajorVessels	Thalassemia	PreviousHeartAttack	StrokeHistory				
## 1	1	Normal		0		0		
## 2	2	Normal		0		0		
## 3	0	Normal		1		0		
## 4	0	Reversible defect		1		0		
## 5	0	Fixed defect		1		0		
## 6	2	Reversible defect		0		0		
##	Residence	EmploymentStatus	MaritalStatus	Outcome				

```
## 1 Suburban Retired Single No Heart Attack
## 2 Suburban Unemployed Married No Heart Attack
## 3 Rural Retired Single Heart Attack
## 4 Suburban Retired Widowed No Heart Attack
## 5 Rural Retired Married Heart Attack
## 6 Rural Unemployed Married Heart Attack
```

```
# Load cleaned data from Status Report #1 and view it (make sure it looks right)
HeartAttackClean <- readRDS("HeartAttackClean.rds")
head(HeartAttackClean)
```

```
## Age Gender Cholesterol BloodPressure HeartRate BMI Smoker Diabetes
## 1 31 0 194 162 71 22.9 0 1
## 2 69 0 208 148 93 33.9 1 1
## 3 34 1 132 161 94 34.0 0 0
## 4 53 0 268 134 91 35.0 0 1
## 5 57 1 203 140 75 30.1 0 1
## 6 41 0 158 154 72 38.7 0 1
## Hypertension FamilyHistory PhysicalActivity AlcoholConsumption Diet
## 1 0 0 6 0 0
## 2 0 0 1 2 0
## 3 1 1 1 3 1
## 4 1 0 6 0 1
## 5 0 0 4 1 2
## 6 0 1 3 2 2
## StressLevel Ethnicity Income EducationLevel Medication ChestPainType
## 1 1 0 64510 0 0 0
## 2 6 1 91773 1 1 1
## 3 3 2 173550 1 1 2
## 4 3 0 43861 0 0 1
## 5 1 0 83404 0 0 0
## 6 4 3 113011 0 0 2
## ECGResults MaxHeartRate ST_Depression ExerciseInducedAngina Slope
## 1 0 173 0.52 0 0
## 2 1 189 3.79 0 1
## 3 2 122 0.17 0 1
## 4 0 104 0.67 0 2
## 5 0 126 5.00 0 2
## 6 1 155 4.30 1 0
## NumberOfMajorVessels Thalassemia PreviousHeartAttack StrokeHistory Residence
## 1 1 0 0 0 0
## 2 2 0 0 0 0
## 3 0 0 1 0 1
## 4 0 1 1 0 0
## 5 0 2 1 0 1
## 6 2 1 0 0 1
## EmploymentStatus MaritalStatus Outcome
## 1 0 0 0
## 2 1 1 0
## 3 0 0 1
## 4 0 2 0
## 5 0 1 1
## 6 1 1 1
```

Logistic Regression Model

```

# Choose predictors
predictors <- c("Age", "Gender", "FamilyHistory", "Ethnicity",
               "Smoker", "BloodPressure", "Cholesterol", "BMI",
               "Diabetes", "Diet", "PhysicalActivity", "HeartRate",
               "AlcoholConsumption")

# Fit logistic regression
fit <- glm(
  reformulate(intersect(predictors, names(HeartAttackClean))), response = "Outcome",
  data = HeartAttackClean,
  family = binomial(),
  na.action = na.omit
)

# Results
summary(fit)           # coefficients (log-odds), p-values

```

```

##
## Call:
## glm(formula = reformulate(intersect(predictors, names(HeartAttackClean))),
##      response = "Outcome", family = binomial(), data = HeartAttackClean,
##      na.action = na.omit)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.750e-02  3.461e-02   0.506  0.6131
## Age          -1.201e-04  2.064e-04  -0.582  0.5608
## Gender         7.578e-03  6.550e-03   1.157  0.2473
## FamilyHistory  1.638e-03  6.550e-03   0.250  0.8025
## Ethnicity     -3.912e-03  2.315e-03  -1.690  0.0911
## Smoker         3.328e-03  6.550e-03   0.508  0.6114
## BloodPressure  3.769e-05  1.260e-04   0.299  0.7649
## Cholesterol    -6.805e-05  5.674e-05  -1.199  0.2304
## BMI            1.336e-04  5.161e-04   0.259  0.7957
## Diabetes       2.905e-03  6.550e-03   0.443  0.6574
## Diet          1.606e-03  4.009e-03   0.401  0.6887
## PhysicalActivity 3.365e-04  1.637e-03   0.205  0.8372
## HeartRate     -1.693e-04  1.888e-04  -0.896  0.3700
## AlcoholConsumption 2.387e-03  2.315e-03   1.031  0.3025
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 517051  on 372973  degrees of freedom
## Residual deviance: 517043  on 372960  degrees of freedom
## AIC: 517071
##
## Number of Fisher Scoring iterations: 3

```

```

exp(coef(fit))           # odds ratios

```

	(Intercept)	Age	Gender	FamilyHistory
	1.0176503	0.9998799	1.0076068	1.0016396

```
##      Ethnicity      Smoker      BloodPressure      Cholesterol
##      0.9960957      1.0033337      1.0000377      0.9999320
##      BMI      Diabetes      Diet      PhysicalActivity
##      1.0001336      1.0029090      1.0016073      1.0003365
##      HeartRate AlcoholConsumption
##      0.9998307      1.0023899
```

```
# Simple in-sample predictions & accuracy (threshold = 0.5)
```

```
p <- predict(fit, type = "response")
pred <- as.integer(p >= 0.5)
table(truth = HeartAttackClean$Outcome, pred = pred)
```

```
##      pred
## truth    0      1
##      0 107199 79459
##      1 106080 80236
```

```
mean(pred == HeartAttackClean$Outcome) # accuracy
```

```
## [1] 0.5025417
```

LDA Model

```
# Choose predictors
```

```
predictors <- c("Age", "Gender", "FamilyHistory", "Ethnicity",
               "Smoker", "BloodPressure", "Cholesterol", "BMI",
               "Diabetes", "Diet", "PhysicalActivity", "HeartRate",
               "AlcoholConsumption")
```

```
# Split into training and test sets
```

```
set.seed(1)
n <- nrow(HeartAttackClean)
id <- sample(n, floor(0.8 * n))
train <- HeartAttackClean[id, ]
test <- HeartAttackClean[-id, ]
```

```
# Fit LDA model
```

```
lda_fit <- lda(Outcome ~ Age + Gender + FamilyHistory + Ethnicity +
              Smoker + BloodPressure + Cholesterol + BMI + Diabetes +
              Diet + PhysicalActivity + HeartRate + AlcoholConsumption,
              data = train)
```

```
# Predict on test data
```

```
lda_pred <- predict(lda_fit, newdata = test)$class
```

```
# Confusion matrix and accuracy
```

```
lda_cm <- table(truth = test$Outcome, pred = lda_pred)
lda_acc <- mean(lda_pred == test$Outcome)
```

```
lda_cm
```

```
##      pred
## truth    0      1
##      0 22303 14971
##      1 22370 14951
```

```
lda_acc
```

```
## [1] 0.4994169
```

QDA Model

```
# Choose predictors
predictors <- c("Age", "Gender", "FamilyHistory", "Ethnicity",
               "Smoker", "BloodPressure", "Cholesterol", "BMI",
               "Diabetes", "Diet", "PhysicalActivity", "HeartRate",
               "AlcoholConsumption")

set.seed(1)
n <- nrow(HeartAttackClean)
id <- sample(n, floor(0.8*n))
train <- HeartAttackClean[id, ]
test <- HeartAttackClean[-id, ]

train <- train[complete.cases(train[, c("Outcome", predictors)]), ]
test_cc <- test[complete.cases(test[, predictors]), ]

qda_fit <- qda(as.formula(paste("Outcome ~", paste(predictors, collapse = " + "))), data = train)
qda_pred <- predict(qda_fit, newdata = test_cc)$class

qda_cm <- table(truth = test_cc$Outcome, pred = qda_pred)
qda_acc <- mean(qda_pred == test_cc$Outcome)

qda_cm
```

```
##      pred
## truth    0    1
##      0 19730 17544
##      1 19698 17623
```

```
qda_acc
```

```
## [1] 0.500744
```

Naive Bayes Model

```
# Choose predictors
predictors <- c("Age", "Gender", "FamilyHistory", "Ethnicity",
               "Smoker", "BloodPressure", "Cholesterol", "BMI",
               "Diabetes", "Diet", "PhysicalActivity", "HeartRate",
               "AlcoholConsumption")

set.seed(1)
n <- nrow(HeartAttackClean)
id <- sample(n, floor(0.8*n))
train <- HeartAttackClean[id, ]
test <- HeartAttackClean[-id, ]

train <- train[complete.cases(train[, c("Outcome", predictors)]), ]
test_cc <- test[complete.cases(test[, predictors]), ]

nb_fit <- naiveBayes(as.formula(paste("Outcome ~", paste(predictors, collapse = " + "))), data = train)
nb_pred <- predict(nb_fit, newdata = test_cc, type = "class")
```

```
nb_cm <- table(truth = test_cc$Outcome, pred = nb_pred)
nb_acc <- mean(nb_pred == test_cc$Outcome)
```

```
nb_cm
```

```
##      pred
## truth    0     1
##      0 22307 14967
##      1 22290 15031
```

```
nb_acc
```

```
## [1] 0.5005429
```

Conclusion:

- All models have around 50% accuracy
- These models are similar to a random sample with an accuracy around 50%
- My goal for next week is to fine tune these models and add more to them to raise the accuracy and make them perfect!